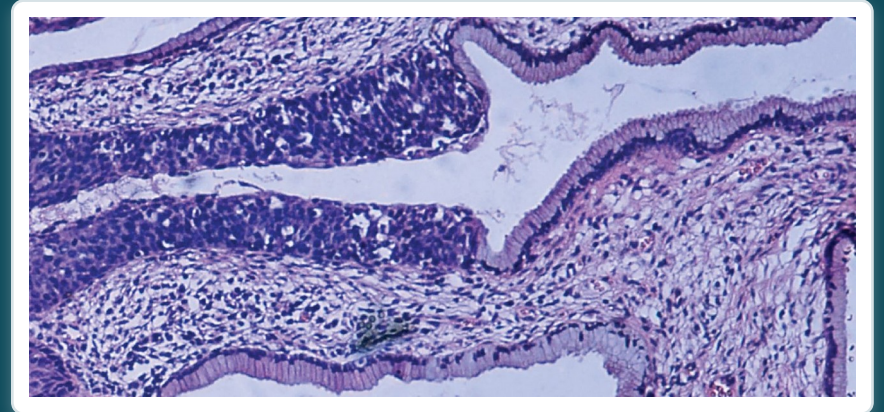


CHAPTER 4

The Tissue Level of Organization

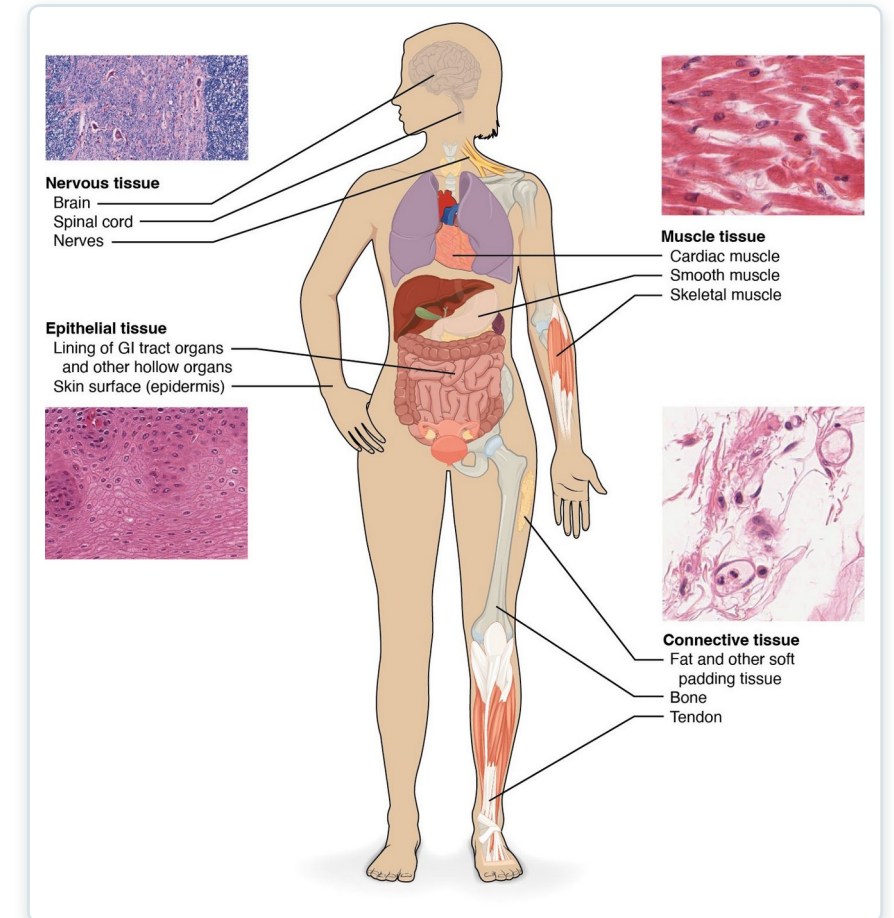
The four primary tissues, their membranes, and how tissues heal.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Identify the main tissue types and discuss their roles in the human body
- 2 Identify the four types of tissue membranes and the characteristics of each that make them functional
- 3 Explain the functions of various epithelial tissues and how their forms enable their functions
- 4 Explain the functions of various connective tissues and how their forms enable their functions
- 5 Describe the characteristics of muscle tissue and how these enable function



Four primary tissue types build every organ.



4.1

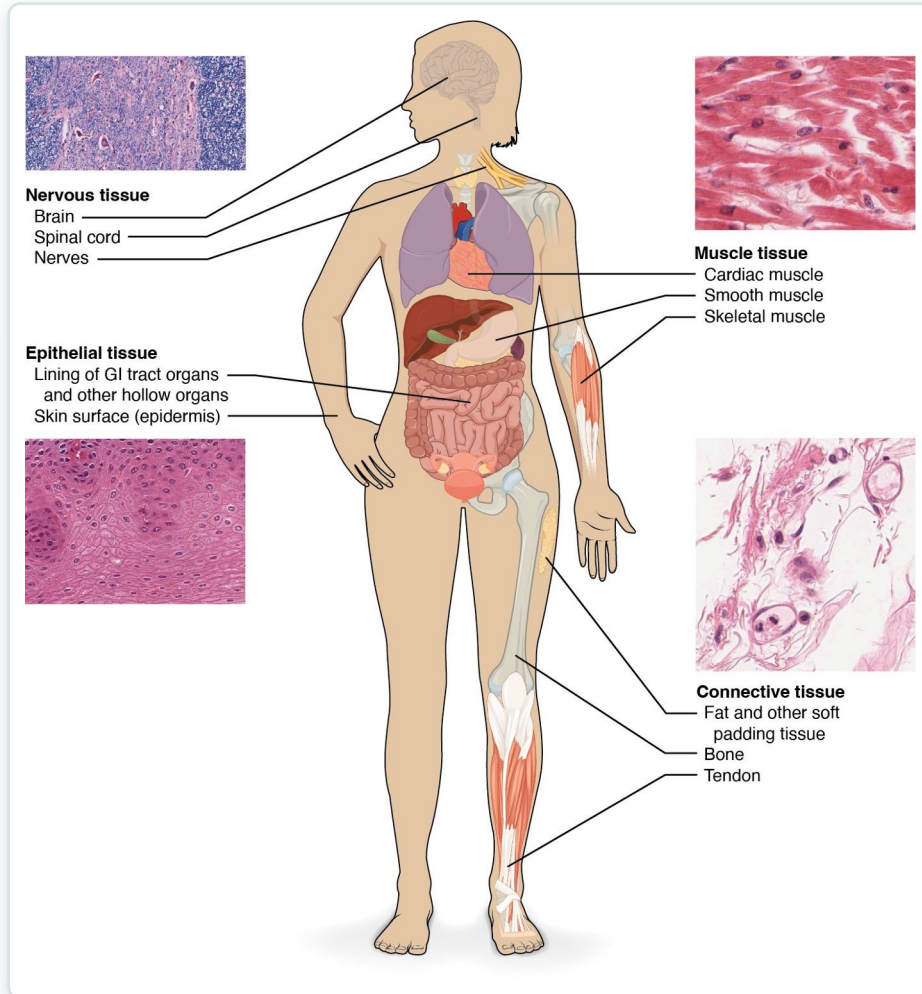
SECTION

Types of Tissues

The four primary tissues and the membranes they form.

SECTION 4.1

The Four Tissue Types

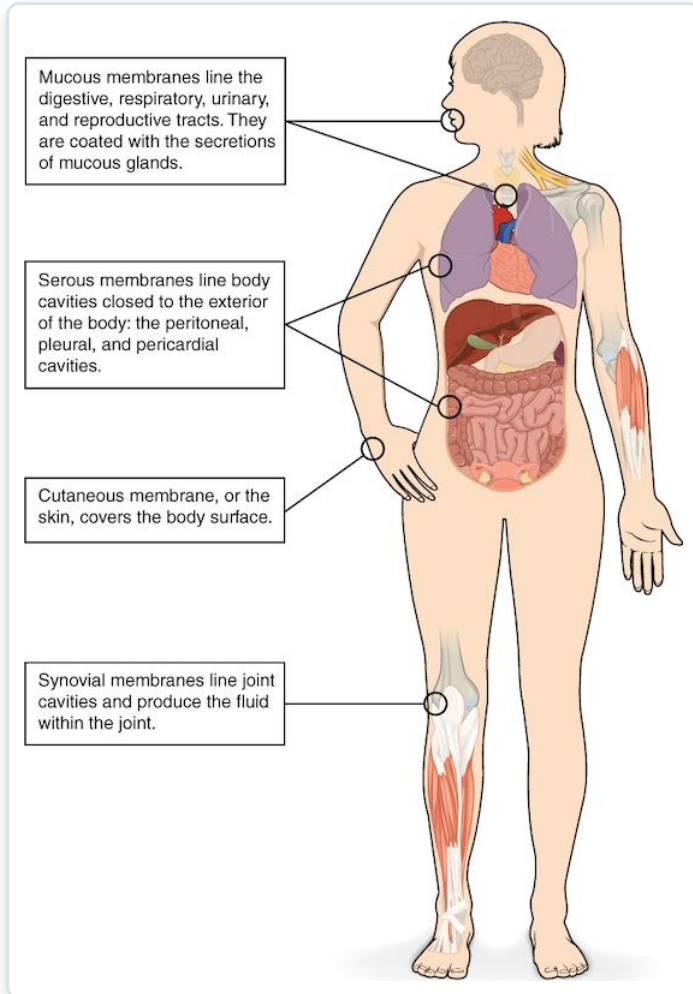


The four primary tissue types

- Epithelial, connective, muscle, nervous
- Tissue = similar cells + their matrix
- Tissues combine to form organs
- Each type has distinct functions

SECTION 4.1

Body Membranes



The body's epithelial membranes

- Membranes line and cover surfaces
- Mucous membranes line passages to the outside
- Serous membranes line closed cavities
- The cutaneous membrane is the skin



4.2

SECTION

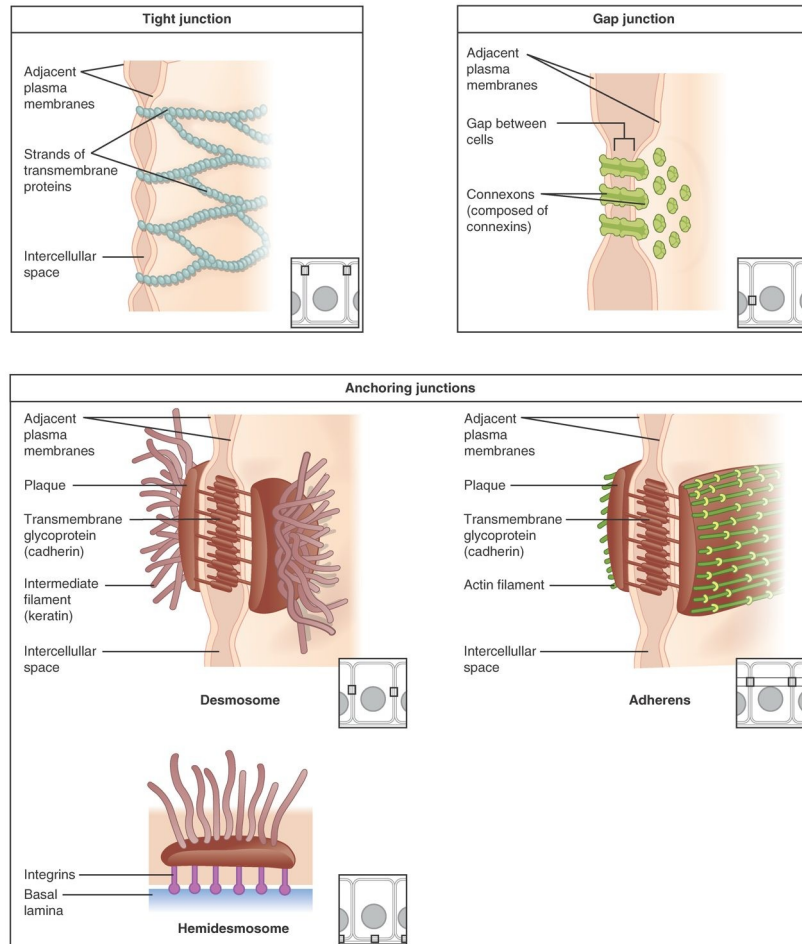
Epithelial Tissue

Covering and lining tissue — classified by cell shape and layering.

SECTION 4.2

Cell Junctions

- Bind epithelial cells together
- Tight junctions seal gaps
- Desmosomes anchor cells
- Gap junctions allow communication

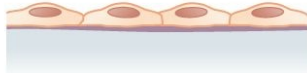
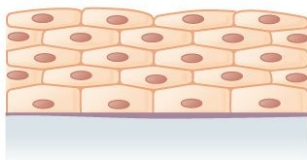
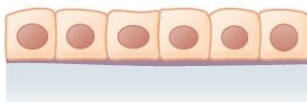
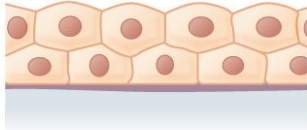
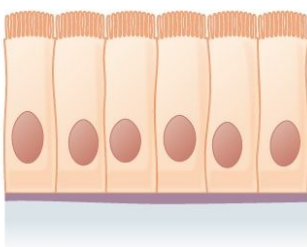
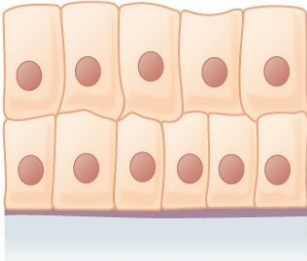
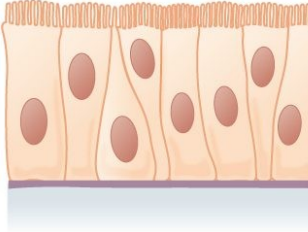


Types of cell junctions

SECTION 4.2

Classifying Epithelium

- Named by cell shape and number of layers
- Shapes: squamous, cuboidal, columnar
- Simple (one layer) vs. stratified (many)
- Structure matches function

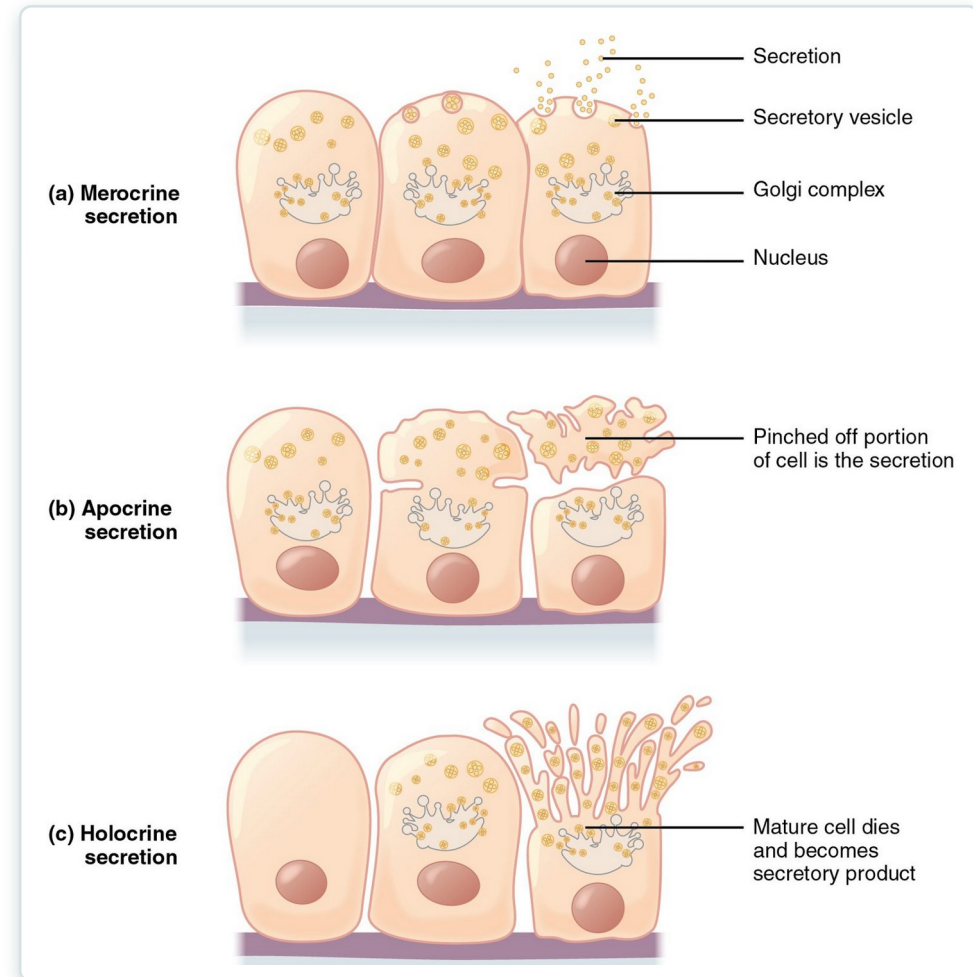
	Simple	Stratified	
Squamous	 Simple squamous epithelium	 Stratified squamous epithelium	
Cuboidal	 Simple cuboidal epithelium	 Stratified cuboidal epithelium	
Columnar	 Simple columnar epithelium	 Stratified columnar epithelium	 Pseudostratified columnar epithelium

Classification of epithelial tissues

SECTION 4.2

Modes of Glandular Secretion

- Exocrine glands secrete onto surfaces
- Merocrine: released by exocytosis
- Apocrine: part of the cell pinches off
- Holocrine: the whole cell ruptures



Merocrine, apocrine, and holocrine secretion



4.3

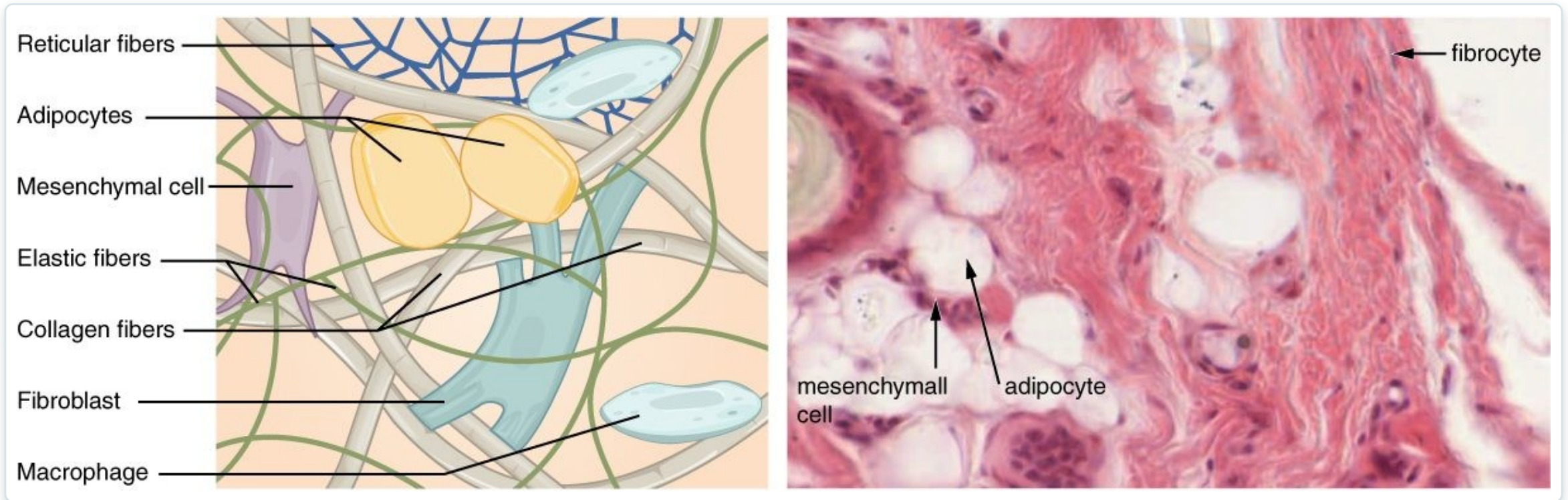
SECTION

Connective Tissue Supports and Protects

The body's most diverse tissue — binding, supporting, and transporting.

SECTION 4.3

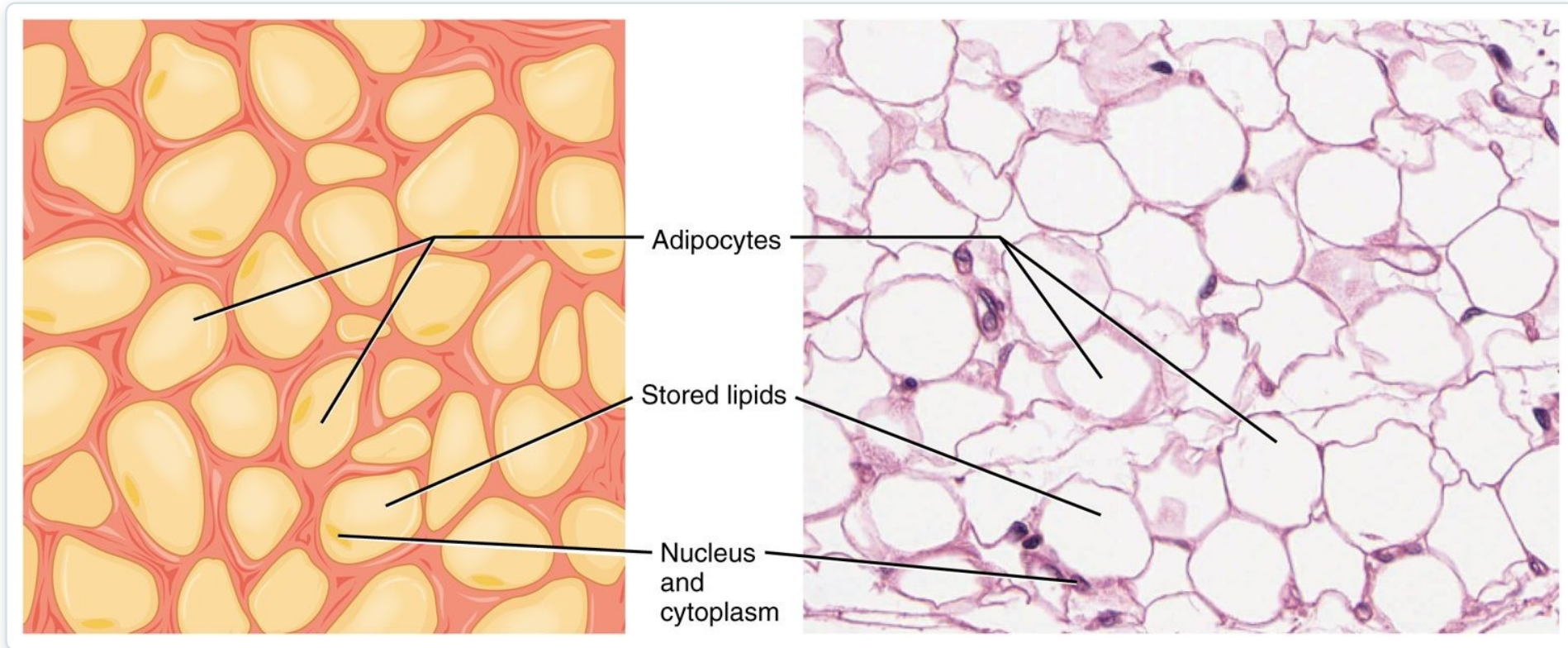
Connective Tissue Overview



Loose connective tissue and its components

- Cells scattered in an extracellular matrix
- Matrix = ground substance + fibers
- Fibroblasts and immune cells reside here
- The most abundant, diverse tissue

Adipose Tissue



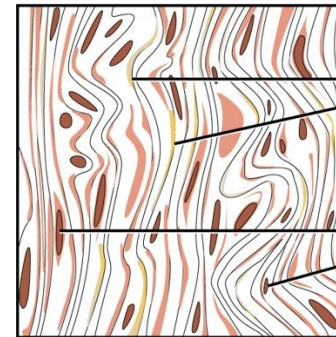
Adipose (fat) tissue

- Stores energy as fat
- Cushions and insulates organs
- Cells are filled with lipid droplets
- An active endocrine tissue

SECTION 4.3

Dense Connective Tissue

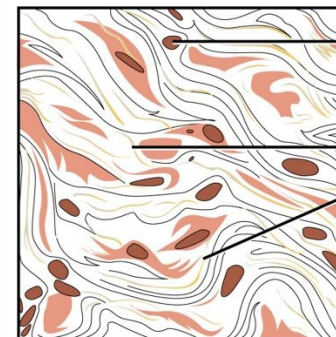
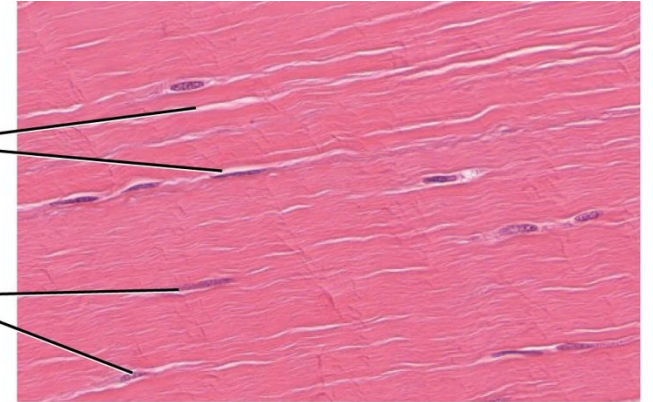
- Packed with collagen fibers
- Dense regular: tendons and ligaments
- Dense irregular: dermis and capsules
- Provides tensile strength



(a) Regular dense

Collagen
fibers

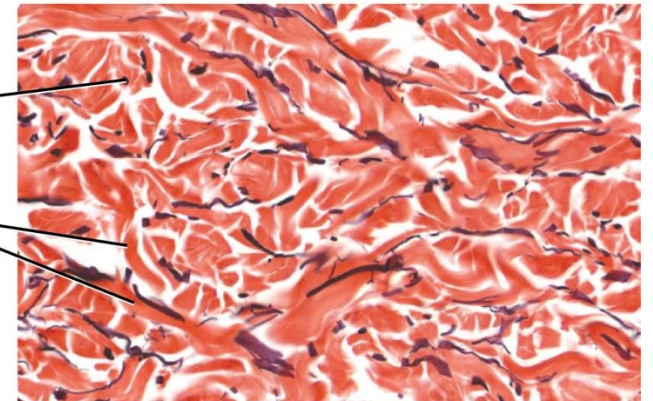
Fibroblast
nuclei



(b) Irregular dense

Fibroblast
nuclei

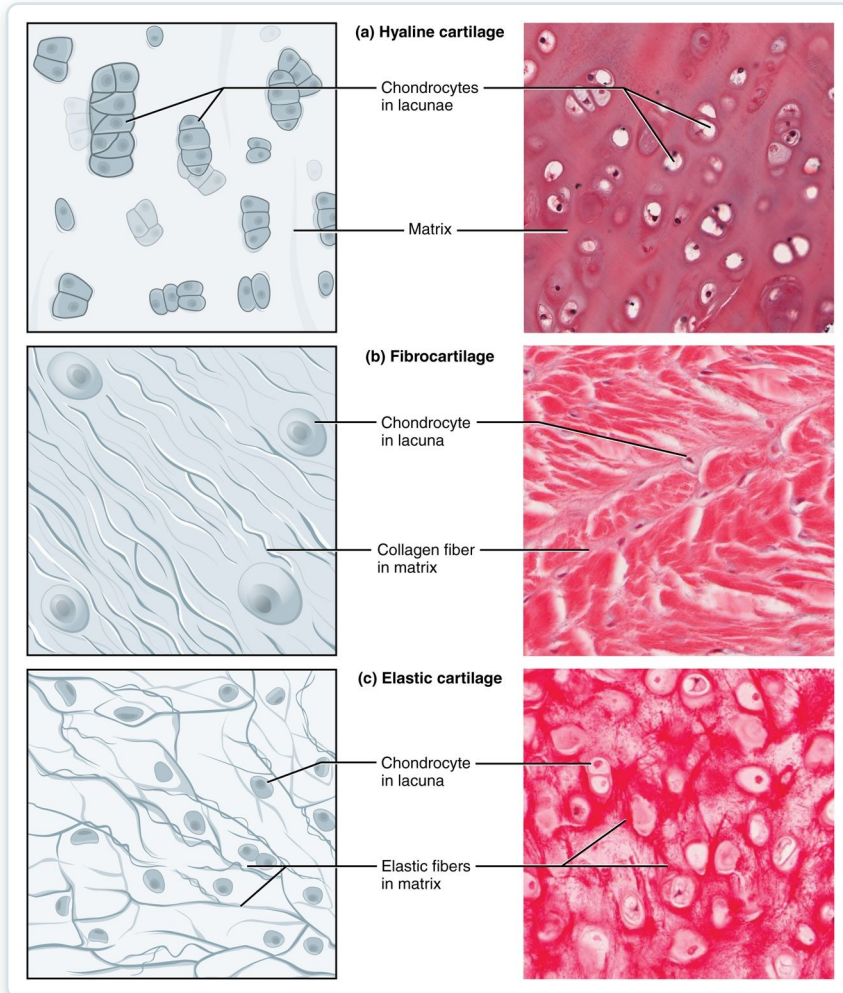
Collagen
fiber
bundles



Dense regular and irregular connective tissue

SECTION 4.3

Cartilage

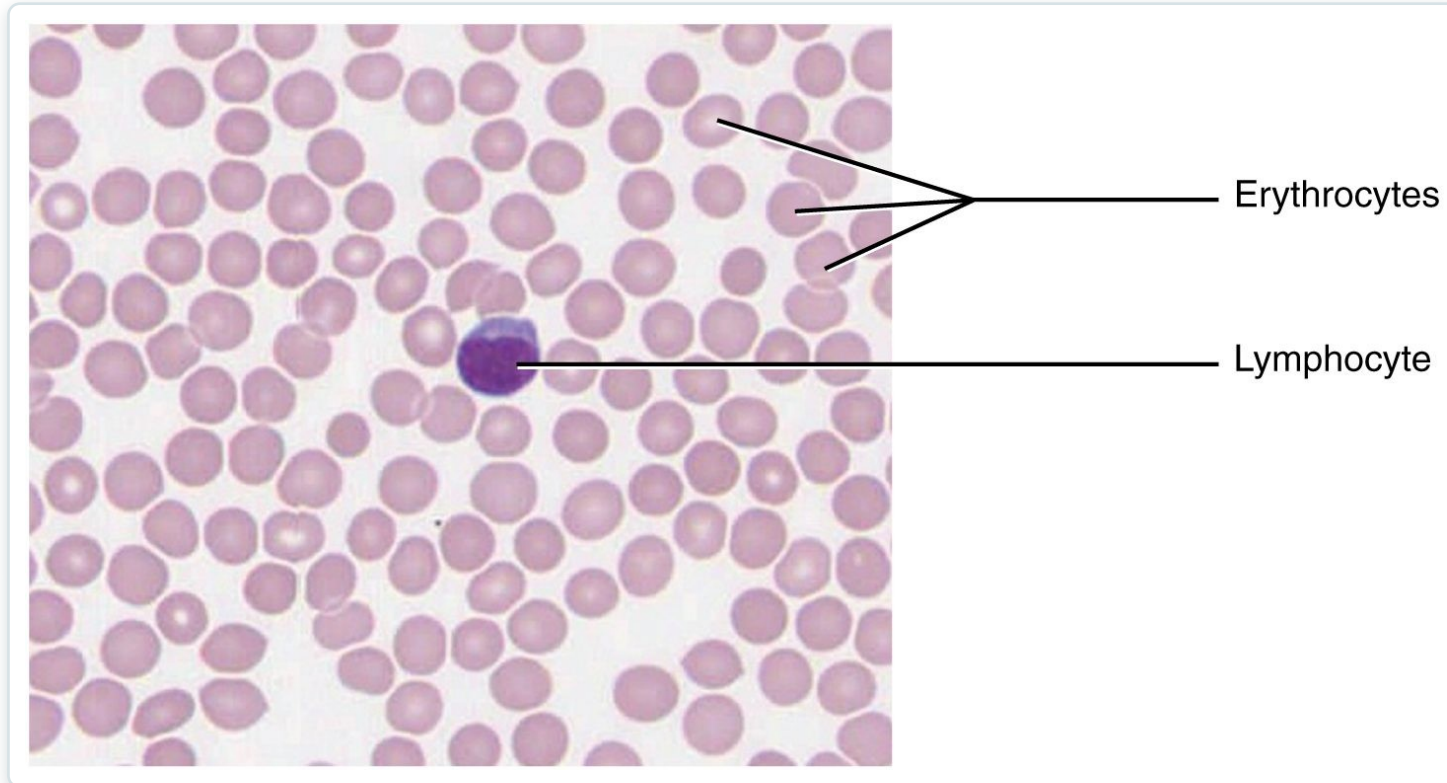


Hyaline, fibrocartilage, and elastic cartilage

- Firm but flexible matrix
- Hyaline: joints, nose, trachea
- Fibrocartilage: discs and menisci
- Elastic: ear and epiglottis

SECTION 4.3

Blood



Blood: erythrocytes and leukocytes

- A fluid connective tissue
- Plasma is the liquid matrix
- Red cells carry oxygen
- White cells defend; platelets clot



4.4

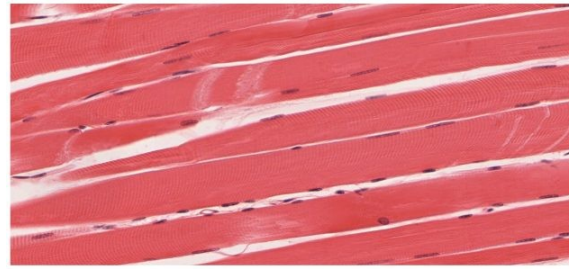
SECTION

Muscle Tissue and Motion

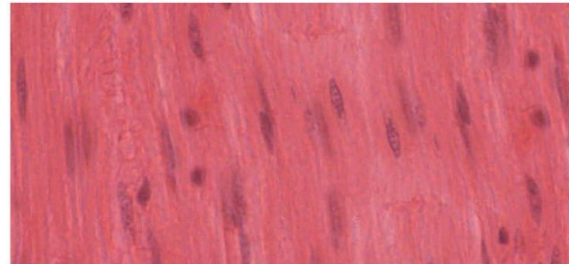
The three muscle types that generate movement.

SECTION 4.4

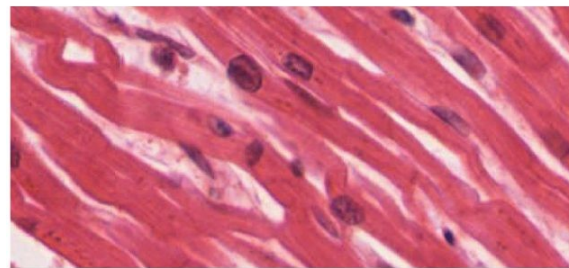
The Three Muscle Tissues



(a)



(b)



(c)

Skeletal, cardiac, and smooth muscle

- Skeletal: voluntary, striated
- Cardiac: involuntary, striated, branched
- Smooth: involuntary, non-striated
- All convert energy into movement



4.5

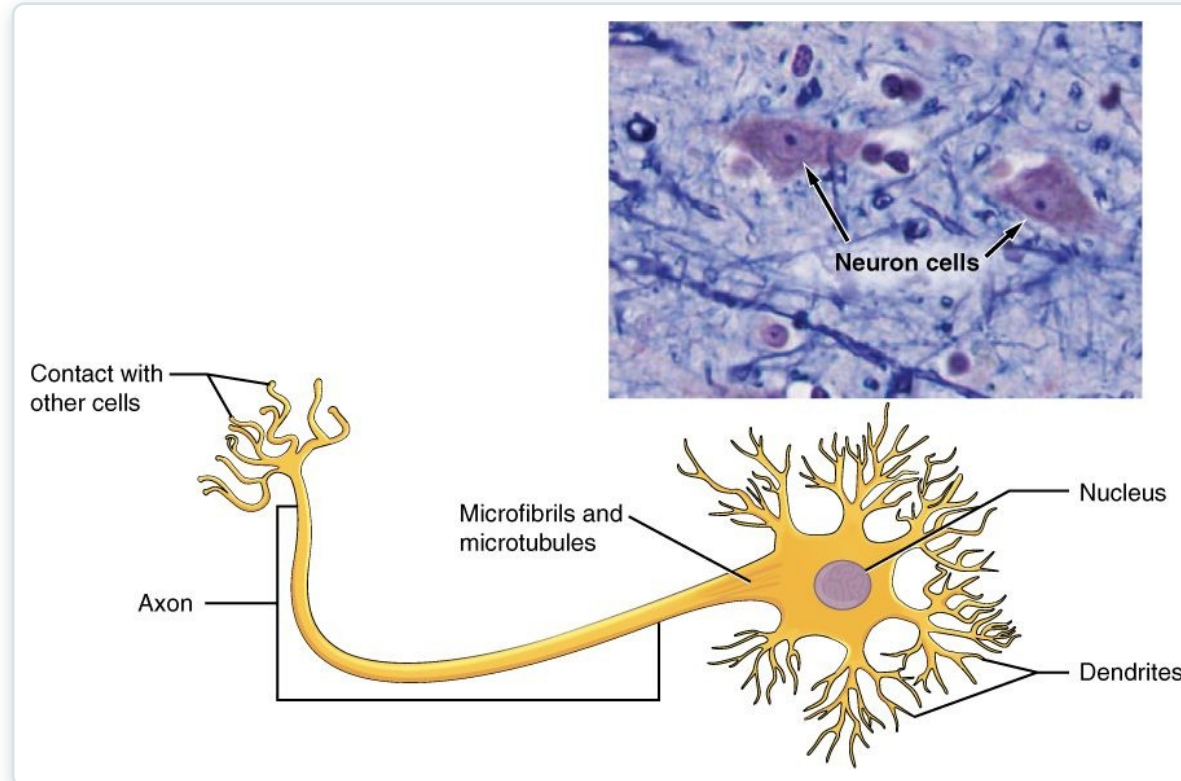
SECTION

Nervous Tissue Mediates Perception and Response

The communication tissue — neurons and their supporting cells.

SECTION 4.5

Nervous Tissue



A neuron of nervous tissue

- Neurons transmit electrical signals
- Cell body, dendrites, and axon
- Dendrites receive; axons send
- Glial cells support neurons



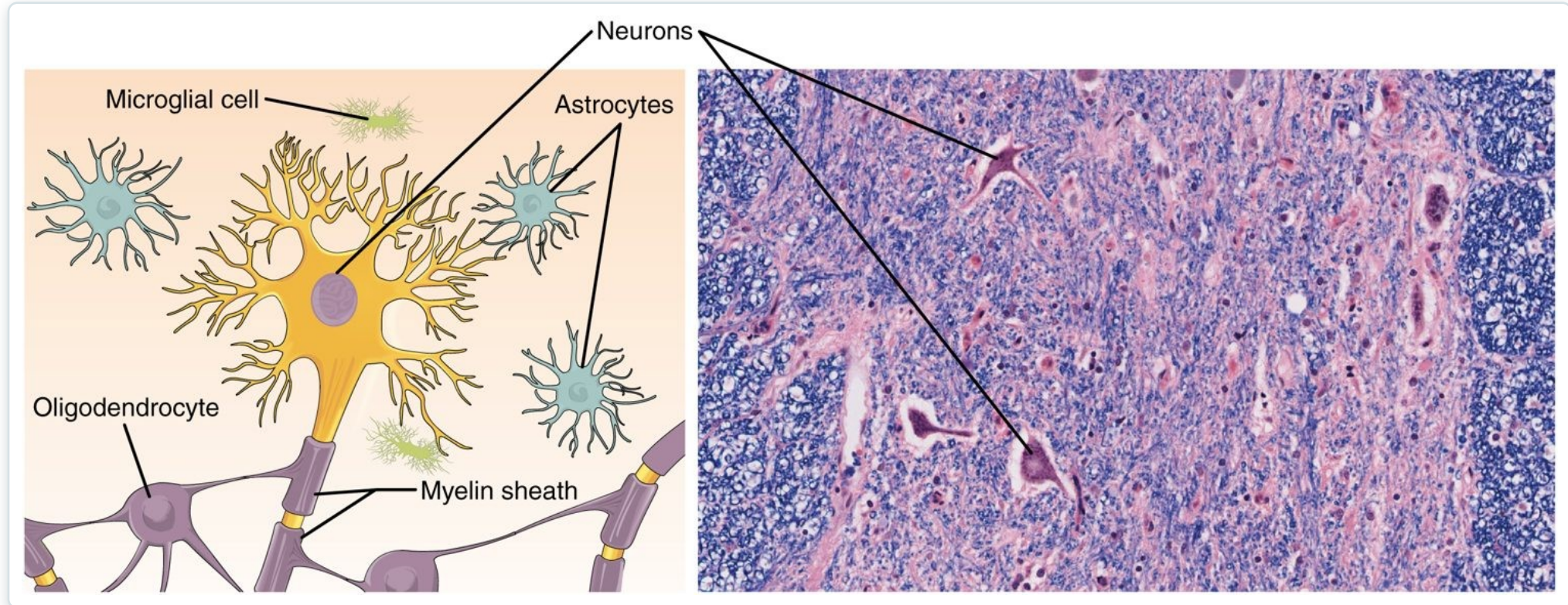
4.6

SECTION

Tissue Injury and Aging

How tissues respond to damage, repair themselves, and change with age.

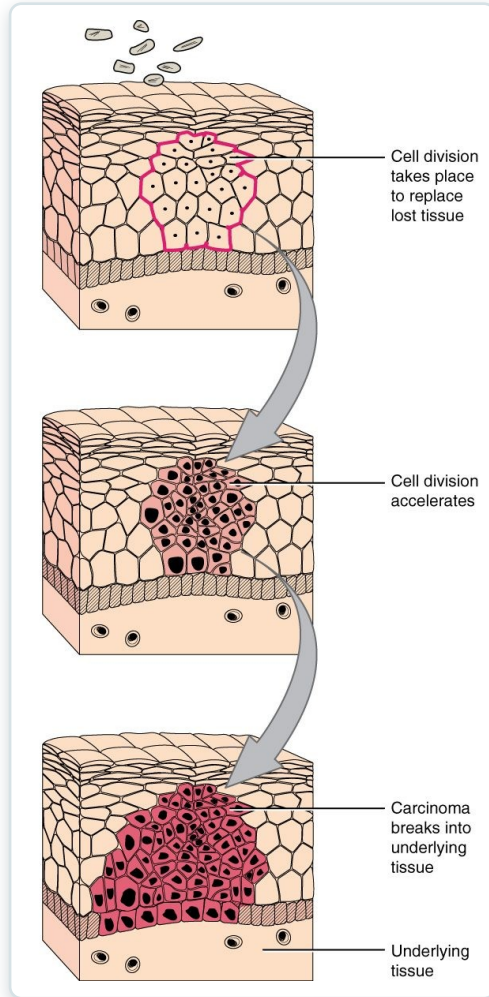
Tissue Injury & Repair



The tissue inflammation and repair process

- Inflammation is the first response
- A clot forms and immune cells arrive
- Granulation tissue rebuilds the area
- Healing by regeneration or scarring

Tissue Aging & Neoplasia



Abnormal cell division and tissue change

- Tissues lose efficiency with age
- Repair slows and tissues thin
- Uncontrolled division → neoplasia
- Malignant tumors invade and spread

Key Takeaways

- Four primary tissues — epithelial, connective, muscle, and nervous — build every organ.
- Epithelium covers and lines, classified by cell shape and layering.
- Connective tissue is the most diverse, defined by its extracellular matrix.
- Muscle generates movement; nervous tissue enables communication.
- Tissues respond to injury through inflammation and repair, and change with age.

Why Tissues Matter in Practical Nursing

Wound Care

Inflammation and repair guide every dressing change and healing assessment.

Skin Integrity

Epithelial membranes are the body's first barrier against infection.

Mobility

The three muscle types underlie movement, heartbeat, and organ function.

Mucous Membranes

Assessing moist linings reveals hydration status and infection.

Cancer Awareness

Uncontrolled cell division produces tumors that invade and spread.

Aging Patients

Slower tissue repair raises the risk of pressure injuries and delayed healing.